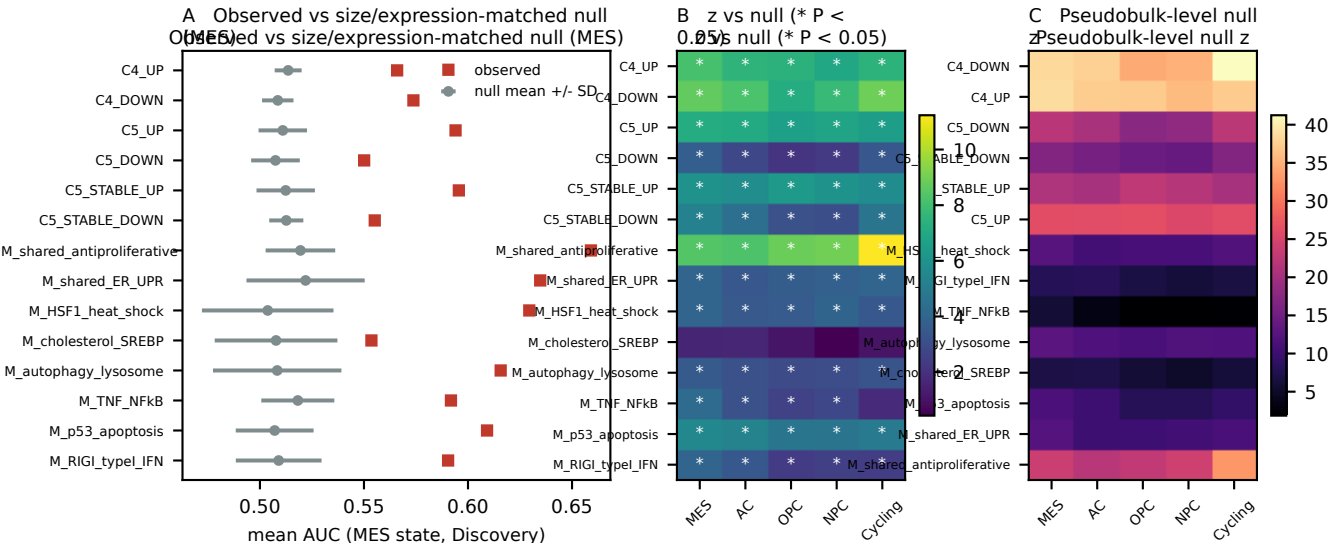




Cell-level panel: 20 size- and expression-matched random gene sets per signature x state (empirical P). Pseudobulk panel: patient-pseudobulk null check. All null controls are Discovery-based and were frozen in Stage C2.